

Descent Lemma

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in C^1$ and ∇f is lipchitz on $\mathbb{R}^n$ with lip. constant $L > 0$.
then

$$f(y) \leq f(x) + \nabla f(x)^T (y-x) + \frac{L}{2} \|y-x\|^2 \quad \forall x, y \in \mathbb{R}^n$$

Generalized descent lemma: Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ & f is L -smooth $\Rightarrow f \in C^1$ and ∇f is lipchitz

Suppose we use gradient descent method on $\min_x f(x)$
then for step length $0 < \alpha \leq \frac{1}{L}$

$$f(x^{k+1}) \leq f(x^k) - \frac{\alpha}{2} \|\nabla f(x^k)\|^2$$

Proof: Use the descent lemma.

$$f(x^{k+1}) \leq f(x^k) + \nabla f(x^k)^T (x^{k+1} - x^k) + \frac{L}{2} \|x^{k+1} - x^k\|^2$$

$$= f(x^k) + \nabla f(x^k)^T (-\alpha \nabla f(x^k)) + \frac{L}{2} \|- \alpha \nabla f(x^k)\|^2$$

$$= f(x^k) - \alpha \|\nabla f(x^k)\|^2 + \frac{\alpha^2 L}{2} \|\nabla f(x^k)\|^2$$

$$\Rightarrow f(x^{k+1}) \leq f(x^k) - \alpha \left(1 - \frac{\alpha L}{2}\right) \|\nabla f(x^k)\|^2$$

$$\left(\text{As } 0 < \alpha \leq \frac{1}{L} \Rightarrow 1 - \frac{\alpha L}{2} \geq \frac{1}{2} \right)$$

$$\therefore f(x^{k+1}) \leq f(x^k) - \frac{\alpha}{2} \|\nabla f(x^k)\|^2$$

Thm: Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$. f is L -smooth. $\min_{x \in \mathbb{R}^n} f(x) > -\infty$

(i.e. $\exists x^* \in \mathbb{R}^n$ at which $\min_{x \in \mathbb{R}^n} f(x) = f(x^*) \rightarrow \text{finite}$)

and let f be convex fnc. Then for $0 < \alpha < \frac{2}{L}$,
 $\lim_{k \rightarrow \infty} f(x^k) = f(x^*)$ in steepest descent method and the convergence is of $O\left(\frac{1}{k}\right)$

Note: we prove this for $\alpha_k = \alpha$ constant $\forall k$.
 $x^{k+1} = x^k - \alpha \nabla f(x^k)$

Proof: we previously got

$$f(x^{k+1}) \leq f(x^k) - \alpha \left(1 - \frac{L\alpha}{2}\right) \|\nabla f(x^k)\|^2$$
$$-f(x^{k+1}) + f(x^k) \geq \alpha \left(1 - \frac{L\alpha}{2}\right) \|\nabla f(x^k)\|^2$$

Sum over $k = 0, 1, 2, \dots, N$

$$f(x^0) - f(x^*) \geq f(x^0) - f(x^{N+1}) \geq \alpha \left(1 - \frac{L\alpha}{2}\right) \sum_{k=0}^N \|\nabla f(x^k)\|^2$$

$\searrow$ $\because f(x^*) \leq f(x^{N+1})$

$$\therefore \underbrace{\alpha \left(1 - \frac{L\alpha}{2}\right)}_{>0} \sum \|\nabla f(x^k)\|^2 \leq f(x^0) - f(x^*)$$

$$\sum \|\nabla f(x^k)\|^2 \leq \frac{f(x^0) - f(x^*)}{\alpha \left(1 - \frac{\alpha L}{2}\right)}$$

$$\sum_{k=0}^N \|\nabla f(x^k)\|^2 \leq \text{constant}$$

Also, the sequence of partial sums of series $\sum_{k=1}^{\infty} \|\nabla f(x^k)\|^2$ is $\sum_{k=0}^{\infty} \|\nabla f(x^k)\|^2$ is an $\uparrow$ seq. bnd. above.
 $\therefore \sum_{k=0}^{\infty} \|\nabla f(x^k)\|^2 < \infty$ (or convergent series)

$$\Rightarrow \lim_{n \rightarrow \infty} \|\nabla f(x^k)\|^2 = 0$$

Next consider

$$\begin{aligned} \|x^{k+1} - x^*\|^2 &= \|x^k - \alpha \nabla f(x^k) - x^*\|^2 \\ &= \|(x^k - x^*) - \alpha \nabla f(x^k)\|^2 \end{aligned}$$

$$= \|x^k - x^*\|^2 + \alpha^2 \|\nabla f(x^k)\|^2 - 2\alpha \nabla f(x^k)^T (x^k - x^*)$$

$$\leq \|x^k - x^*\|^2 + \alpha^2 \|\nabla f(x^k)\|^2 - 2\alpha (f(x^k) - f(x^*))$$

$$\left(\because f(x^*) - f(x^k) \geq \nabla f(x^k)^T (x^k - x^*) \rightarrow \text{convex} \right)$$

$$\Rightarrow f(x^k) - f(x^*) \leq \frac{1}{2\alpha} \|x^k - x^*\|^2 - \frac{1}{2\alpha} \|x^{k+1} - x^*\|^2 + \frac{\alpha}{2} \|\nabla f(x^k)\|^2$$

$$\text{Gen. descent lemma} \Rightarrow f(x^{k+1}) \leq f(x^k) - \frac{\alpha}{2} \|\nabla f(x^k)\|^2$$

$$\therefore \cancel{f(x^k)} - f(x^*) \leq \frac{1}{2\alpha} \|x^k - x^*\|^2 - \frac{1}{2\alpha} \|x^{k+1} - x^*\|^2 + \cancel{f(x^k) - f(x^{k+1})}$$

$$\Rightarrow f(x^{k+1}) - f(x^*) \leq \frac{1}{2\alpha} \|x^k - x^*\|^2 - \frac{1}{2\alpha} \|x^{k+1} - x^*\|^2$$

Sum it over $k = 0, 1, \dots, N$

$$\sum (f(x^{k+1}) - f(x^*)) \leq \frac{1}{2\alpha} \left[\|x^0 - x^*\|^2 - \|x^{N+1} - x^*\|^2 \right]$$

$$\leq \frac{\|x^0 - x^*\|^2}{2\alpha}$$

In steepest descent

$$f(x^{k+1}) \leq f(x^k) \quad \forall k$$

$$\therefore (N+1) (f(x^{N+1}) - f(x^*)) \leq \frac{\|x^0 - x^*\|^2}{2\alpha}$$

$$\therefore 0 \leq f(x^{N+1}) - f(x^*) \leq \frac{\|x^0 - x^*\|^2}{2\alpha (N+1)} = O\left(\frac{1}{N}\right)$$

$\therefore$ By sandwich thm

$$\lim_{N \rightarrow \infty}$$

$$f(x^{N+1}) \rightarrow f(x^*)$$

$$\text{with } O\left(\frac{1}{N}\right)$$

$$* \rightarrow f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$f: C^1$$

∇f is lipchitz

f is convex

$$0 < \alpha < \frac{1}{L} < \frac{2}{L}$$

SDM converges

with order $\left(\frac{1}{N}\right)$

N : # of iterations

linear cgs.

Geometrically: $f(x) = \frac{1}{2} x^T Q x - b^T x$

Exact line search

$$\alpha = \frac{d^T d}{d^T Q d}$$

$$d = -\nabla f(x)$$

$$\alpha_k \rightarrow \min_{\alpha > 0} \phi(\alpha) = f(x^k + \alpha d^k)$$

$$\phi'(\alpha_k) = 0$$

$$(d^k)^T \nabla f(x^k + \alpha_k d^k) = 0$$

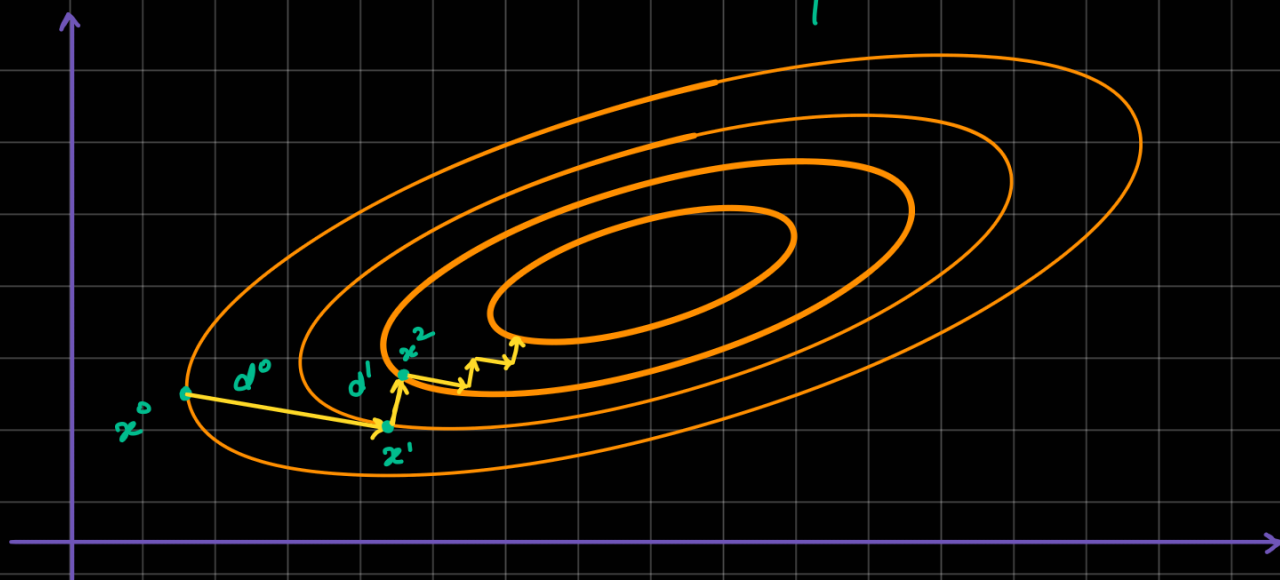
$$d^{k^T} \nabla f(x^{k+1}) = 0$$

(chain rule)

$$\Rightarrow d^{kT} d^{k+1} = 0$$

Two consecutive directions in SDM are orthogonal (or numerically close to 90°)

For quad. fnc, the level curves $f(x) = c$ for $c > 0$ $c \in \mathbb{R}$ are concentric ellipse.



$x^0 \rightarrow$ initial pt
 $d^0 \rightarrow -\nabla f(x^0)$

Everytime we move in a zig-zag path which leads to a slowly convergence specially in elongated ellipses.

* Nesterov Accelerated Gradient Descent : (AGD)

- Idea came from Poljak heavy ball method.

$$x^{k+1} = x^k - \alpha \nabla f(x^k) + \underbrace{\beta (x^k - x^{k-1})}_{\text{momentum term}}$$

$\rightarrow$ convergence is still linear

$$x^{k+1} = y^k - \alpha \nabla f(x^k)$$

Nesterov

0

$$y^k = x^k + \beta (x^k - x^{k-1})$$

momentum
term

$$x^{k+1} = y^k - \alpha \nabla f(y^k)$$

Lookahead Gradient